IM233AI

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1.1	Illustrate the Pyramidal Structure of the work.	02	2	1
1.2	Distinguish between internal element and external element in cycle time analysis.	02	4	1
1.3	Enumerate the categories of worker machine systems based on the levels of operator attention required.	02	1	2
1.4	Differentiate between snap back and cumulative timing methods.	02	4	2
1.5	What is lean production?	02	2	3
1.6	Why is an allowance added to the normal time to complete the standard time?	02	2	2
1.7	Identify the broad categories of factors causing fatigue in workers.	02	2	4
1.8	List any four techniques of Work Measurement.	02	1	2
1.9	In the context of lean manufacturing what is production leveling?	02	2	3
1.10	In lean manufacturing, _____ refers to waste or non-value-adding activities, while _____ refers to overburdening people or equipment.	02	2	3

PART-B

a	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	08	4	1
b	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The *PFD* allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8-hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8-hour shift that corresponds to the *PFD* allowance factor of 16%.	08	3	2
a	Review the applicability of three categories of Charting / diagrammatic techniques used by industrial engineers for collecting, displaying and analyzing data on a given work system.	08	2	1
b	Provide an overview on any four techniques of Methods Engineering.	08	2	1

		OR		
4	a	Enumerate and provide a brief note on the principles of motion economy.		
	b	Review the phases involved in the systematic approach that can be used to accomplish a methods engineering study.		
5	a	What is a Work Element? Mention at least six guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.		
	b	The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a, b, d, e and f are operator controlled and external to machine cycle and they are performance rated at 90%. If the machine allowance is 12% and worker *PFD* allowance is 8%, determine the normal time and standard time for the work cycle.		

Element	a	b	c	d	e	f
Observed Time(min)	0.12	0.28	0.78	1.04	1.45	1.80

		OR		
6	a	What is meant by *PMTS*? Distinguish between First level *PMTS* and Higher level *PMTS* on the basis of at least five characteristics.	08	
	b	Analyse the causes for variability in the observed work element times that occur from one work cycle to another work cycle.	08	
7	a	Enumerate the seven wastes as advocated by Taiichi Ohno in Lean Production philosophy. Explain any four in detail.	10	
	b	Distinguish between Push System of production control and Pull System of production control. Support your answer with an appropriate illustration.	06	
		OR		
8	a	In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.	08	
	b	Explain the role of Total Productive Maintenance (*TPM*) in automation with examples.	08	
9	a	Employees in an office environment report eye strain, fatigue, and discomfort. Analyse the physical work environment and recommend improvements.	08	
	b	What is ergonomics and explain its purpose in production systems. Discuss how ergonomics and human factors are connected to engineering.		
		OR		
10	a	Highlight on the concept of anthropometry and its importance in designing for the human body, with examples.		
	b	Describer how mechanics can be applied to the human body to enhance safety and efficiency in the workplace.		

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Supplementary Examinations Oct/Nov-2024

Industrial Engineering and Management

WORK SYSTEMS DESIGN

3 Hours *Maximum Marks: 100*

tions to candidates:

Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
.1	Illustrate pyramidal structure of work.	02	2	1
.2	What is meant by the term productivity?	02	2	1
.3	List the symbols used in traditional industrial engineering charting techniques.	02	1	2
.4	List any two objectives of methods engineering.	02	1	2
.5	Mention the elements of standard time for manual work.	02	1	3
.6	Expand the acronym MOST.	02	1	3
.7	What is meant by Autonomation?	02	2	1
.8	Mention the main activities involved in 5s.	02	1	2
.9	What is meant by the term Anthropometry?	02	1	4
.10	Distinguish between the terms: 'Accident' and 'Hazard'.	02	2	4

PART-B

	During the base year in a mill 326000 tons of steel were produced using 203,000 labor hours. In the next year, the output was 341,000 tonns using 246,000 labor hours. Determine the (i) Labor productivity ratio for the base year, (ii) Labor productivity ratio for the second year, (iii) Labor productivity index for the second year.	10	3	1
	Categorize the excess non-productive activities. Write a brief note on each of the category.	06	3	1
	Enumerate the steps in systematic approach of methods engineering application for an improvement.	10	2	2
	Evaluation of alternative improvement proposals is done considering two categories of features and specifications. Write a note on them.	06	2	2
	OR			
	Review the basic data collection tools and analysis techniques used for problem solving.	10	4	2
	Illustrate and elaborate on automation migration strategy.	06	3	2
	Classify the methods to determine time standards. Support your explanation with an illustration for better clarity.	10	3	3
	Illustrate/sketch the model indicating the factors that must be standardized before a standard time can be set for a task.	06	3	3

OR

6	a	Enumerate the steps involved in direct time study procedure to arrive at standard time for a job/activity.
	b	A regular cycle of work consists of three elements a, b, c. Element 'd' is irregular element performed every five cycles. Using the following details; compute normal time and standard time for the work cycle, using an allowance factor of 15%.

Work element	a	b	c	d
Observed time (min)	0.56	0.25	0.50	1.10
Performance Rating (%)	100	80	110	100

7	a	Summarize the seven forms of 'waste' that needs to be eliminated as advocated by Taiichi Ohno to make the production systems lean.
	b	What is meant by the term 'Kaizen'? How is Kaizen implemented for achieving the goal of continuous improvement in lean systems?

OR

8	a	Distinguish between 'push' and 'pull' systems of production control. Support your explanation with an illustrative example.
	b	Illustrate the structure of a lean production system and highlight the key elements.
9	a	Sketch the block diagram model of the interactions in a human-machine system. Provide a brief note on different elements in the system.
	b	Write a note on the primary casual factors for the occurrence of industrial accidents.

OR

10	a	Review the three broad topical areas of Ergonomics as a field of study.
	b	Distinguish between 'Single incident traumas' and 'Cumulative trauma disorders' in the context of Ergonomics.

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III Semester B. E. Examinations April/May-2024

Industrial Engineering and Management

WORK SYSTEMS DESIGN

e: 03 Hours *Maximum Marks: 100*

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1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
	Illustrate the pyramidal structure of work.	02	2	1
	Define the term "Productivity".	02	1	1
	Distinguish between "Methods Design" and "Methods Analysis".	02	2	2
	One of the approaches for dealing with the automation projects in methods engineering is USA principle. Expand the acronym 'USA'.	02	1	2
	What is meant by application speed ratio in the context of work measurement?	02	2	3
	Expand the acronym *'MOST'*.	02	1	3
	What is the principal objective of *JIT*?	02	2	2
	What does the term 'Jidoka' refer to?	02	1	2
	List the four primary variables that influence the climate in the work place environment.	02	1	4
)	Distinguish between the terms, "industrial accident" and "hazard".	02	2	4

PART-B

	Distinguish between the terms, 'basic productive work content' and 'excess non-productive work content.' Also illustrate with two examples for each category of excess non-productive activities that add to total task time.	08	3	1
	A work group of 5 workers in a certain month produced 500 units of output working 8 hrs/day for 22 days in a month. Suppose in the next month, the some work group produced 600 units but were only 20 work days in the month. Using labor productivity measure, determine the productivity index using the prior month as a base.	08	4	1
	Enumerate the steps in a systematic approach that can be used to accomplish a methods engineering study.	08	2	2
	Summarize the features and applicability of the following tools used by industrial engineers for basic data collection and analysis as part of problem solving process.			
	i) Defect concentration diagram			
	ii) Scatter diagram	08	3	2

OR

4	a	Enumerate any eight principles of motion economy related to the use of human body.	08
	b	Illustrate the automation migration strategy and explain the three phases included in the typical strategy followed by manufacturing organizations.	08
5	a	Depict a model indicating the factors that must be standardized before a standard time can be set for a task. Also explain the elements of the model.	08
	b	The observed element times and performance ratings collected in a direct time study are indicated in the table below. The snapback/flyback timing method was used. The *PFD* allowance is 14%. All the elements are regular elements in the work cycle. Determine: i) Normal time and ii) The standard time for the cycle.	08

Work Element	*a*	*b*	*c*	*d*
Observed time (min)	0.22	0.41	0.30	0.37
Performance Rating (%)	90	120	100	90

OR

6	a	Examine the three sequence (activity sequence) models in Basic MOST and indicate the standard sequence of model parameters for each.	08
	b	The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements a, b, c and d are performed by the operator and element 'm' is a machine element; Element 'b' is an internal element performed by the operator simultaneously with element 'm'; and element 'd' is an irregular element performed once every 15 cycles. Observed times and performance ratings are given in the table. The *PFD* allowance factor is 15%, and the machine allowance is 20%. Determine: i) Normal time and ii) Standard time for the work cycle.	08

Worker element	*a*	*b*	*c*	*d*
Observed time in minutes – manual	0.22	0.65	0.47	0.75
Performance rating (%)	100	80	100	100
Machine element		*m*		
Observed time in minutes – machine	*idle*	1.56	*idle*	*idle*

7	a	Illustrate the structure of a lean production system and explain in detail any two ingredients of the system.	08
	b	Examine the three prerequisites for successful implementation of Just-in-Time production system.	08

OR

8	a	Examine any four aspects influencing the stable and reliable production operations in a manufacturing environment.	08
	b	Motivated workers participation in continuous improvement is critical for achieving the objective of lean production system. In the above context, discuss the relevance of visual management and 5s.	08

	Depict the model of interactions in a typical human machine system. Provide an overview of the human components of interest in the system.	08	3	4
	Work is performed in an environment that includes both physical and social aspects. Classify the broad categories into which physical environment can be segregated from ergonomics perspective. Elaborate on any two categories in detail.	08	4	4
	OR			
	Elucidate the three factors commonly identified in the ergonomics literature as the primary reasons for occurrence of industrial accidents.	08	2	4
	Describe any four common cummulative trauma and other occupational disorders in a chosen ssector of your interest.	08	4	4